CMPG211

Week 7 Assignment: Objects and text files - Properties

Write a program for an estate agency to manage the letting of properties. Your solution must implement the concepts of **encapsulation**, **inheritance** and **polymorphism**.

Information required on **all the properties** listed to be rented:

- type of property (residential (R) or business (B)),
- name of the owner,
- contact number of owner,
- duration of the contract (in months).

Residential properties:

Type of residence ((F) Flat, (T) Townhouse or (H) House), number of bedrooms, number of bathrooms, number of garages, rent per month.

The rent for residential properties depend on the type of property and the number of bedrooms.

Flat – R8000pm + R 500pm per bedroom
Townhouse – R 15000pm + R 1000pm per bedroom
House – R 12000pm + R800pm per bedroom

Business properties:

Size in square meters, insurance per month, rent per month.

The rent for business properties is calculated based on the size of the property. The rent per square meter is R250.00

Two separate reports must be displayed for the two types of properties.
The output for **residential properties** must be displayed as follows:

Owner	Contact number	Months	Type	BedR	BathR	Rent(pm)
Joe Bloom	0879678223	24	T	4	3	R19000

The output for **business properties** must be displayed as follows:

Owner	Contact number	Months	Size	Insurance(pm)	Rent(pm)
Sue Smith	0723456872	12	280	R7000.00	R70000.00

1. Write a superclass **Property**.
   - Identify the common properties for a property and declare them as private **instance fields.**

   - **Constructors**:
     No arguments constructor (also called the **default** constructor)
     Parameterized constructor.

   - **Methods**:
     - Accessor methods (getters) for the instance fields.

     - A **toString()** method to return a string that will neatly display all the information to describes the object. Example:

       Joe Bloom 0879678223 2

2. Write a **subclass** called **ResProperty** that inherits from the **Property** superclass

   - Declare local instance field(s).

   - Two **constructors**:
     No arguments constructor
     Parameterized constructor:

   - **Methods**:
     - Accessor methods for the instance fields

- **calcMonthRent()** method to calculate and return the monthly rent.

- The **toString()** method must return a string that will neatly display all the information to describes the object.

  Example:

  Joe Bloom 0879678223 24 T 4 3 R19000

3. Write the **subclass** called **BussProperty** that inherits from the **Property** superclass

- Declare the local instance fields.

- Two **constructors**:
  No arguments constructor
  Parameterized constructor:

- **Methods**:
  - Accessor methods for the instance fields.

  - A method **calcMonthRent()** to calculate and return the monthly rent.

  - A method **calcMonthInsurance()** to calculate and return the monthly insurance amount.

  - A **toString()** method must return a string that will neatly display all the information to describes the object.

  Example:

  Sue Smith 0723456872 12 280 R7000.00 R70000.00

4. Create a **FileClass** class and write a method to read information from a text file called ***properties.txt*** into the an ArrayList.

   Example of the content of the text file:

```
R#Joe Bloom#0879678223#24#T#4#3
R#Sam Xaba#0849558202#12#F#2#1
B#Sue Smith#0723456872#12#280
R#Lee West#0826458288#12#H#3#2
B#John Pule#0723456872#24#320
B#Abe Thiem#0843412573#6#330
R#Craig Moore#0721236285#18#T#3#2
```

   Write a **get method** to return the ArrayList.

   Write a method called **writeTownHToFile()** to write the name and cell phone number of the owner and the rent per month of all the townhouses to a text file called ***townhouses.txt***
   Example of content of the file:

```
Joe Bloom    0879678223   R19000.00
Craig Moore  0721236285   R18000.00
```

5. Write an Interface class called Propable for the method to calculate the rent amount.

6. Write a **testProperty** class:

   Declare an **ArrayLists** of type Propable.

   Write **two static methods** to display the information of the properties as two separate lists – one method for residential and one method for business properties.
   The methods must receive the ArrayList to display.
   Display suitable headings and subheadings.
   Display the rent amount for the properties as well.

Call the methods from the main method.

Example of output:

```
Residential Properties
Name            Cell num        Months     Type     Beds     Baths        Rent(pm)
Joe Bloom       0879678223      24         T        4          3          R19000.00
Sam Xaba        0849558202      12         F        2          1          R9000.00
Lee West        0826458288      12         H        3          2          R14400.00
Craig Moore     0721236285      18         T        3          2          R18000.00

Business Properties
Name            Cell num        Months     Size    Insurance    Rent(pm)
Sue Smith       0723456872      12         280      R7000.00    R70000.00
John Pule       0723456872      24         320      R8000.00    R80000.00
Abe Thiem       0843412573      6          330      R8250.00    R82500.00
```

Call the method **writeTownHToFile()** to write the information about townhouses to a text file.